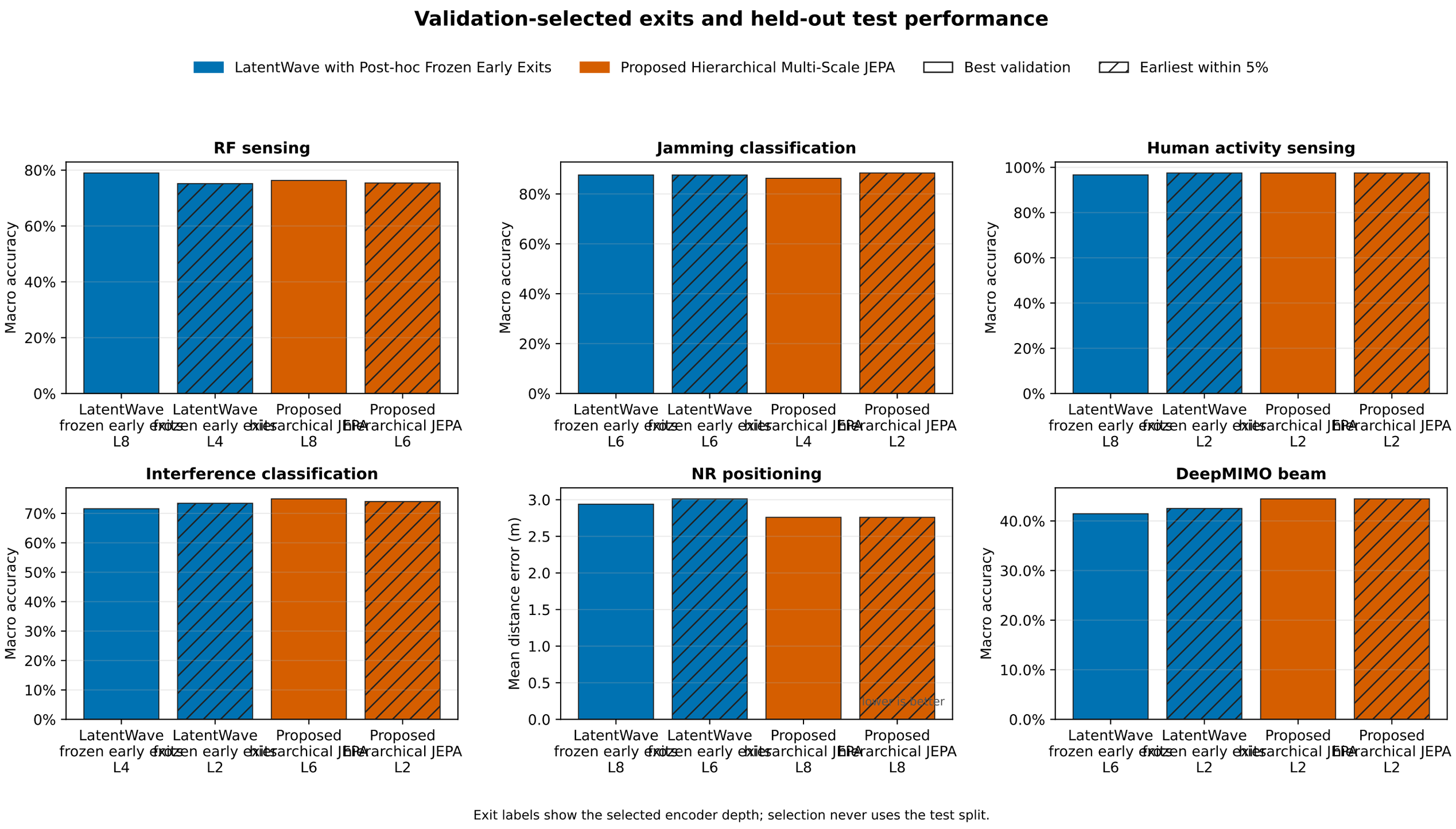
Validation-selected exits and held-out test performance
LatentWave with Post-hoc Frozen Early Exits
Proposed Hierarchical Multi-Scale JEPA
Best validation
Earliest within 5%
RF sensing
Macro accuracy
0%
20%
40%
60%
80%
LatentWave frozen early exits L8
LatentWave frozen early exits L4
Proposed hierarchical JEPA L8
Proposed hierarchical JEPA L6
Jamming classification
Macro accuracy
0%
20%
40%
60%
80%
LatentWave frozen early exits L6
LatentWave frozen early exits L6
Proposed hierarchical JEPA L4
Proposed hierarchical JEPA L2
Human activity sensing
Macro accuracy
0%
20%
40%
60%
80%
100%
LatentWave frozen early exits L8
LatentWave frozen early exits L2
Proposed hierarchical JEPA L2
Proposed hierarchical JEPA L2
Interference classification
Macro accuracy
0%
10%
20%
30%
40%
50%
60%
70%
LatentWave frozen early exits L4
LatentWave frozen early exits L2
Proposed hierarchical JEPA L6
Proposed hierarchical JEPA L2
NR positioning
Mean distance error (m)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
lower is better
LatentWave frozen early exits L8
LatentWave frozen early exits L6
Proposed hierarchical JEPA L8
Proposed hierarchical JEPA L8
DeepMIMO beam
Macro accuracy
0.0%
10.0%
20.0%
30.0%
40.0%
LatentWave frozen early exits L6
LatentWave frozen early exits L2
Proposed hierarchical JEPA L2
Proposed hierarchical JEPA L2
Exit labels show the selected encoder depth; selection never uses the test split.